

Comparing the Maynard–Tao and Green–Tao Programs: Correlations, Tools, Frontiers, and Prospects for Unification

Executive summary

The bounded-gaps program driven by James Maynard ¹ and Terence Tao ², and the long-arithmetic-progressions program initiated by Ben Green ³ and Tao, can be compared cleanly through the lens of **correlation order** and the **type of pseudorandomness required**. The bounded-gaps line is fundamentally a **two-point / second-moment** problem: it seeks (infinitely often) prime pairs (or clusters) within bounded diameter using **optimized Selberg/GPY-type sieve weights** plus **distribution of primes in arithmetic progressions**, typically at Bombieri–Vinogradov strength or its refinements. ⁴

The long-progressions line is a **k-point / higher-order uniformity** problem: it seeks prime configurations governed by multilinear averages (e.g., the counting operator for k-term APs), requiring control of higher uniformity norms (Gowers) and an understanding of the structured obstructions captured by **nilsequences**. Its core mechanism is **transference** (a relative Szemerédi theorem) after a **W-trick** removes local congruence obstructions. ⁵

A partial “unified framework” exists at the level of **weights and majorants**: Green–Tao explicitly builds a pseudorandom majorant for primes using divisor-sum/sieve ideas traceable to the same Selberg/GPY ecosystem that powers bounded gaps, while Maynard–Tao can be interpreted as an extremal optimization of such weights for detecting *at least two primes* among shifts. ⁶

However, a **single grand machine** that simultaneously (i) optimizes sieve weights to attack *binary* patterns (twin-like) and (ii) proves *higher-order* equidistribution against all structured obstructions (nilsequences) does not currently exist. The best-known bridges are in **short-interval regimes** and in **distribution-to-large-moduli with factorable weights**, where analytic number theory advances feed sieve optimization and where higher-order uniformity on average becomes available for multiplicative functions. ⁷

Quantitatively, the bounded-gaps record certified in the literature remains the Polymath8b bound $H_1 \leq 246$ (unconditional) and $H_1 \leq 6$ under generalized Elliott–Halberstam-type hypotheses. ⁸ A 2025 manuscript circulated on ResearchGate claims an unconditional improvement to $H \leq 234$, but (as of the sources retrieved here) it is not traceable to a standard archival venue (e.g., arXiv/peer-reviewed journal), so it should be treated as **unverified** until independently checked. ⁹

On the Green–Tao side, the central qualitative result is the existence of arbitrarily long APs in the primes (Annals 2008), and the modern endpoint is a toolkit that combines: (i) transference, (ii) inverse theorems for Gowers norms, and (iii) Möbius orthogonality to nilsequences, enabling Hardy–Littlewood-type asymptotics for wide classes of *finite-complexity, non-degenerate* linear systems. ¹⁰

Program goals and correlation regimes

Core goals

Bounded gaps / prime clusters. Define $H_1 := \liminf_{n \rightarrow \infty} (p_{n+1} - p_n)$. The twin prime conjecture is $H_1 = 2$. The Maynard-Tao breakthrough (and improvements by Polymath8b) establish that $H_1 < \infty$ with explicit bounds, and more generally that for every $m \geq 1$,

$$H_m := \liminf_{n \rightarrow \infty} (p_{n+m} - p_n) < \infty,$$

i.e. infinitely many bounded-length intervals contain at least $m + 1$ primes. 11

Long arithmetic progressions / general linear patterns. For each k , prove existence of primes $a, a + d, \dots, a + (k - 1)d$ all prime. Green-Tao proves this for all k ; later work treats many other linear systems in primes (under “finite complexity” and non-degeneracy conditions). 12

Correlation order: “2-point” versus “k-point”

A useful unifying language is to view primes via a proxy like the von Mangoldt function Λ and ask about **averages of products of shifts**.

Bounded gaps is essentially 2-point (second-moment) detection. A stylized bounded-gaps criterion is: choose a finite admissible set $\mathcal{H} = \{h_1, \dots, h_k\}$ and a nonnegative weight $w(n)$. If one can show, for large x ,

$$\sum_{x \leq n \leq 2x} w(n) \sum_{i=1}^k \theta(n + h_i) > (m - 1) \log x \sum_{x \leq n \leq 2x} w(n),$$

then some n has at least m primes among $n + h_i$. This compares a “weighted prime count” to a “weighted mass” and is powered by controlling sums of Λ (or θ) in arithmetic progressions with moduli coming from the weight’s divisor structure. 13

Even though k may be large, the analytic engine is still dominated by **quadratic functionals in the weights** (optimizing a second-moment ratio), and by **1-point distributional input** for primes in progressions sufficient to evaluate those weighted sums. 14

Long APs are truly k-point (higher-order) correlation. Counting 3-term APs already requires controlling the bilinear average

$$\sum_{n,d} \Lambda(n) \Lambda(n + d) \Lambda(n + 2d),$$

and k -term APs require k -fold products. The natural norms governing these counts are the Gowers U^{k-1} norms: small U^{k-1} implies near-random behavior for k -term AP counting operators, while large U^{k-1} indicates correlation with structured objects (nilsequences of step $k - 2$). 15

Key comparison point. Green-Tao's later "linear equations in primes" program explicitly excludes the most "binary" degenerate cases (e.g., twin primes), because affine dependencies force a different (harder) correlation regime. The 2006/2010 "Linear equations in primes" preprint/paper frames this as excluding "binary" cases like twin primes and Goldbach. ¹⁶

Method pipelines and principal technical inputs

Visual aid: milestone timeline

Events below are documented across the primary papers and retrospectives cited in the surrounding text: Green-Tao (2004/2008), Zhang (2014), Maynard (2015), Polymath8b (2014), Matomäki-Radziwiłł (2016), higher-uniformity short-interval work (2020 preprint → 2023 Annals), and the short-interval bounded-gaps extension (2017). ¹⁷

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timeline
  title Prime-patterns timeline (bounded gaps vs long APs)
  2004 : Long APs in primes (preprint)
  2008 : Long APs in primes (Annals publication)
  2010 : Linear equations in primes (Annals)
  2012 : Inverse theorem for Gowers norms (Annals)
  2012 : Möbius orthogonal to nilsequences (Annals)
  2013 : First bounded-gaps breakthrough (gap < 70 million)
  2014 : Bounded gaps optimized to 246 (unconditional)
  2016 : Multiplicative functions in short intervals (Annals)
  2017 : Bounded gaps in short intervals (extension)
  2020 : Higher-order uniformity in short intervals (preprint)
  2023 : Higher-order uniformity in short intervals (Annals)
  2025 : Claimed improvement of 246 to 234 (unverified)

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Visual aid: side-by-side pipelines

The pipelines correspond to the canonical presentations in the Annals papers and Polymath8b's overview, and to Tao's research-thread blog posts summarizing the projects and later uniformity developments. ¹⁸

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flowchart LR
  subgraph BG["Bounded-gaps pipeline (sieve optimization)"]
    A["Choose admissible k-tuple H"] --> B["Build Selberg/GPY-type weight w(n)"]
    B --> C["Estimate weighted prime sum via primes-in-AP input"]
    C --> D["Optimize a quadratic ratio over weights"]
    D --> E["Positivity forces ≥2 primes among shifts infinitely often"]
    E --> F["Extract explicit H bounds (e.g., 246)"]
  end

  subgraph AP["Long-AP pipeline (transference + higher uniformity)"]
    G["W-trick: pre-sieve small primes (n=Wm+b)"] --> H["Majorant v dominating

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primes"]
  H --> I["Verify pseudorandomness: linear forms / Gowers norms"]
  I --> J["Inverse theorem: structure = nilsequence correlation"]
  J --> K["Relative Szemerédi/transference"]
  K --> L["Deduce k-term APs in primes for all k"]
end

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Main tools and typical inputs

The following is a “researcher’s checklist” view: what each program *actually* needs in order to close estimates.

Bounded gaps: Selberg/GPY/Maynard sieve with distributional input.

- Weight design: nonnegative weights $w(n)$ built from truncated divisor sums (Selberg sieve / GPY framework), with Maynard’s refinement introducing more flexible multi-dimensional coefficient choices and a clean variational optimization problem. ¹⁹
- Prime distribution input: at minimum, **Bombieri–Vinogradov-level** distribution of primes in arithmetic progressions is enough to prove bounded gaps (qualitatively) and obtain explicit constants; refinements (Zhang/Polymath8a-type) improve constants and extend to more delicate regimes. ²⁰
- Admissible tuples: existence and diameter of admissible k -tuples is an explicit computational component that influences the final numeric bound. ²¹
- Limitation: Polymath8b explicitly frames a “parity problem” limitation—pure sieve methods cannot by themselves force a specific even gap (e.g., 2) without deeper distributional breakthroughs. ⁸

Long APs: transference + higher-order Fourier analysis (Gowers/nilsequences).

- W-trick: take $W = \prod_{p \leq w} p$ (often small primorials like 30 or 210 in expository contexts) and restrict to residue classes coprime to W to eliminate local congruence obstructions before measuring pseudorandomness. ²²
- Majorant construction: embed primes into a pseudorandom “almost prime” set (or a majorant measure ν) using sieve/divisor-sum technology; the original Green–Tao preprint identifies this as a key ingredient via Goldston–Yıldırım ideas. ²²
- Transference principle: prove a **relative Szemerédi theorem**—if ν is pseudorandom enough, then any subset of ν of positive *relative* density contains long APs. ²³
- Structure theory and obstructions: if a function fails Gowers uniformity, it correlates with an appropriate nilsequence (inverse theorem). This is formalized by Green–Tao–Ziegler’s inverse theorem for $U^{s+1}[N]$. ²⁴
- Möbius randomness against structured objects: Green–Tao’s “Möbius is strongly orthogonal to nilsequences” gives a powerful *anti-structure* statement, central to controlling error terms in prime-pattern counting via nilsequence decomposition. ²⁵

Quantitative results and representative theorems

Bounded gaps: landmark theorems and best certified constants

The canonical quantitative arc is:

- Yitang Zhang ²⁶ proved bounded gaps between consecutive primes with an explicit bound $H_1 < 7 \times 10^7$ (Annals 2014). ²⁷
- Maynard proved $H_1 \leq 600$ and, crucially, $H_m < \infty$ for all m (Annals 2015; preprint 2013), by refining GPY sieve weights and the associated optimization problem. ²⁸
- The Polymath8 collaboration’s “b” component (D.H.J. Polymath) pushed the unconditional bound to $H_1 \leq 246$ and obtained stronger conditional bounds such as $H_1 \leq 6$ assuming a generalized Elliott–Halberstam hypothesis; it also states explicit asymptotic bounds for H_m and discusses sieve-theoretic limitations. ²⁹

It is often useful to record the *growth-form* bounds for H_m . The Polymath8b abstract reports Maynard’s $H_m \ll m^3 e^{4m}$ and conditional $H_m \ll m^3 e^{2m}$ (EH), and then reports Polymath improvements such as $H_m \ll m e^{(4 - \frac{24}{181})m}$ unconditional (and $H_m \ll m e^{2m}$ under EH). ⁸

Short-interval bounded gaps. A significant modern refinement is that bounded-gap phenomena can be forced *inside short intervals*: Ryan Alweiss ³⁰ and Sammy Luo ³¹ prove a Maynard–Tao-style bounded-gaps statement in intervals $[x - x^\delta, x]$ for $\delta \in [0.525, 1]$, leveraging short-interval prime-density tools in the style of Baker–Harman–Pintz and Maynard–Tao. ³²

Surveys / later perspective. Maynard’s survey “Gaps between primes” provides a post-breakthrough map of weak prime k -tuple conjectures and how small/large gaps fit together. ³³ A high-level account of the “distribution input” frontier is also visible in Maynard’s later work on primes in AP to large moduli with well-factorable weights. ³⁴

Claimed improvement to 234. A 2025 manuscript titled “A Weighted Distribution of Primes and a New Unconditional Bound on Gaps Between Primes” claims $H \leq 234$ (improving 246) and advertises a “well-factorable weight function” innovation, but the only located record here is via ResearchGate; treat this as provisional until it appears in an archival venue and/or is independently verified. ⁹

Long APs and the “modern Green–Tao endpoint”

Green–Tao theorem (Annals 2008). The primes contain arithmetic progressions of length k for every k . The proof is described as combining Szemerédi’s theorem, a transference principle (relative Szemerédi), and a sieve-theoretic embedding of primes into a pseudorandom measure. ²³

From APs to general linear patterns. “Linear equations in primes” (Annals 2010) develops an asymptotic theory for many prime solutions to systems of affine-linear forms of finite complexity, but explicitly excludes “binary” degenerate cases such as twin primes; this paper is part of the program that becomes unconditional once the relevant inverse-theorem and Möbius/nilsequence components are established. ³⁵

Structure theorems that complete the program. * The Green-Tao-Ziegler inverse theorem for the Gowers $U^{s+1}[N]$ norm gives the precise structured obstructions: large Gowers norm implies correlation with an s -step nilsequence. ²⁴

* Green-Tao's "Möbius is strongly orthogonal to nilsequences" settles the Möbius-nilsequence conjecture with strong decay (log-power savings), eliminating a major structural obstruction in the analytic number theory side of the transference program. ²⁵

Quantitative modernizations. On the purely additive-combinatorics side, optimizing quantitative dependence has been a major theme. Frederick Manners ³⁶ gives a new proof of the inverse theorem with explicit (though still very large) quantitative bounds, avoiding regularity/nonstandard tools in that setting. ³⁷ A Bourbaki exposé by Bloom surveys these quantitative inverse results and their role. ³⁸

Modern frontiers and where the two programs already touch

Frontier themes in bounded gaps

Better distribution in arithmetic progressions, in weighted form. A striking post-2015 direction is to go beyond Bombieri-Vinogradov in *special averaged/weighted senses*, notably for well-factorable weights and moduli as large as $x^{3/5-\epsilon}$. Maynard's "Primes in arithmetic progressions to large moduli II: well-factorable estimates" is a representative primary source, and is explicitly framed as having consequences for "level of distribution for sieve weights." ³⁹

This matters because the Maynard-Tao/Polymath8b sieve outputs are constrained by how far one can push distributional estimates for primes against the moduli created by the sieve weights.

Short-interval phenomena. Bounded gaps in short intervals (Alweiss-Luo) show that "global" bounded gaps can be refined into "local" windows when combined with short-interval prime-density technology. This directly creates a meeting point with the "local uniformity" agenda on the Green-Tao side. ⁴⁰

Frontier themes in Green-Tao / higher-order uniformity

Local (short-interval) higher-order uniformity on average. The 2023 Annals paper by Kaisa Matomäki ⁴¹, Maksym Radziwiłł ⁴², Tao, Joni Teräväinen ⁴³, and Tamar Ziegler ⁴⁴ proves higher-order uniformity for bounded multiplicative functions in short intervals **on average**, upgrading earlier "local Fourier uniformity" to higher degrees and even to nilsequence tests. ⁴⁵

This paper's arXiv abstract explicitly links the result to **inverse theory for Gowers norms** and to nilsequence correlation tests. ⁴⁶

Classical short-interval groundwork. The foundational Annals result "Multiplicative functions in short intervals" (2016) established powerful cancellation properties for Möbius/Liouville on almost all short intervals, representing a milestone on the "local randomness" axis. ⁴⁷

Methodological consolidation. Tao's expository posts, for instance on "higher uniformity ... in short intervals" and the earlier Möbius/nilsequences conjecture, document how these strands fit the longer transference agenda. ⁴⁸

Existing overlap: “sieve weights as majorants” and “local-to-global”

A concrete overlap is that Green–Tao’s original construction embeds primes inside a pseudorandom superset using sieve/divisor-sum weights (as highlighted already in the 2004 preprint abstract), while bounded gaps is literally an optimization of sieve weights to force prime-rich shifts. ⁶

A second overlap is the modern pivot toward **short intervals**: bounded gaps can be forced in $[x - x^\delta, x]$ for $\delta \geq 0.525$, while multiplicative functions (and nilsequence correlations) can be controlled on average in short intervals for a broad range of lengths. ⁴⁹

This suggests a realistic “bridge substrate”: local information, averaged over x , where both sieve and higher-uniformity tools have traction.

Toward a unified framework: three hybrid approaches (feasibility, partial results, obstacles)

Higher-order sieve weights

Idea. Replace the classical “detect at least two primes among shifts” objective with weights designed to detect k -point patterns (e.g., 3-term APs) directly, e.g. optimizing a weight $w(n, d)$ for the multilinear form

$$\sum_{n,d} w(n, d) \Lambda(n) \Lambda(n + d) \Lambda(n + 2d).$$

This would be a real unification: a sieve-style extremal problem tailored for an additive configuration-counting operator.

Known partial analogues. * Green–Tao already uses a sieve majorant ν to make configuration counting feasible via transference. ²³

* Maynard–Tao already solves a sophisticated variational optimization for second-moment-type objectives, showing the power of optimization-guided weight design. ¹³

Main obstacles. * Evaluating such higher-order weighted sums demands control of prime correlations beyond what BV-type input provides. The “linear equations in primes” program achieves many such counts only after importing higher-order structure theory (Gowers/nilsequences) and excluding degenerate binary cases; attempting to do this *purely* by sieve runs into parity and correlation barriers. ⁵⁰

* Polymath8b explicitly argues that certain strengthened conclusions are the “best possible” from purely sieve-theoretic considerations (parity-type limitations), indicating that an unadorned sieve generalization is unlikely to reach twin-prime-level specificity, let alone fully structured k -point asymptotics. ⁸

Feasibility assessment. Methodologically plausible as a research program (it defines a concrete optimization object), but its success is bottlenecked by unavailable multi-linear distribution estimates for primes unless it imports substantial higher-order harmonic analysis—at which point it begins to resemble Green–Tao more than Maynard–Tao.

Embedding nilsequence control into sieve optimization

Idea. Use nilsequence decompositions/inverse theorems to isolate “structured” components of the arithmetic objects arising in sieve evaluations (error terms, remainder terms, distribution in progressions), and then optimize sieve weights in a regime where structured obstructions are provably absent or controlled.

Known partial results and evidence. * The inverse theorem for Gowers norms precisely characterizes the obstructions to uniformity as nilsequence correlations. ²⁴

Möbius orthogonality to nilsequences provides a strong way to kill structured obstructions arising from multiplicative randomness. ⁵¹

The modern short-interval higher-uniformity results explicitly replace polynomial-phase tests by nilsequence tests, showing that nilsequence control can be made local and quantitative on average. ⁵²

Main obstacles. * The sieve weights in bounded gaps are not Möbius/Liouville; they are engineered divisor-sum objects whose correlations can themselves have structured components. Understanding these in nilsequence language is nontrivial and would require a careful “structure theorem for sieve weights,” not currently standard.

* The bounded-gaps bottleneck is often in pushing distribution of primes in APs to larger moduli (or stronger levels), where the relevant objects are exponential sums and bilinear forms in primes rather than nilsequence correlations per se. Maynard’s distribution-to-large-moduli work is framed in “well-factorable weights” language, not nilsequence language. ³⁹

Feasibility assessment. Conceptually promising (it targets the exact obstruction class that higher-order Fourier analysis identifies), but currently lacks a mature “interface layer” translating sieve-weight optimization problems into nilsequence-structured/uniform decompositions in a way that improves numeric bounds like H_1 .

Short-interval uniformity as a bridge substrate

Idea. Work in windows $[x, x + H]$ (or $[x - x^\delta, x]$) and average over x . In this regime, both camps have genuine theorems: bounded gaps can be forced in short intervals, and higher-order uniformity of multiplicative functions (and nilsequence correlations) can be controlled on average.

Known partial results. * Bounded gaps in $[x - x^\delta, x]$ for $\delta \geq 0.525$ (Alweiss–Luo) provides a bounded-gap statement **localized** in short intervals. ⁴⁰

* Higher uniformity of multiplicative functions in short intervals on average (Annals 2023) provides exactly the kind of local high-order pseudorandomness control that Green–Tao-type counting requires, but locally. ⁴⁵

Main obstacles. * Translating “multiplicative function uniformity” into “prime uniformity” remains delicate: primes are controlled via von Mangoldt weights and sieve majorants, and local uniformity results are typically proved for bounded multiplicative functions (e.g., Liouville) rather than Λ itself. ⁵³

* Short-interval prime distribution itself is a deep topic; bounded gaps in short intervals uses heavy short-interval PNT technology (BHP-type), and improvements in the δ threshold are likely constrained by that analytic input. ³²

Feasibility assessment. This is the most concrete and currently active bridge: it aligns with demonstrated theorems on both sides and offers a shared set of testable conjectures (“local pseudorandomness implies local patterns”) that can be attacked with existing toolchains.

Comparative table and recommended research experiments

Comparative table of attributes

Attribute	Bounded-gaps program (Maynard–Tao / Polymath8b lineage)	Long-AP / linear-pattern program (Green–Tao lineage)
Central problem	Prove $H_1 < \infty$ and sharpen explicit bounds; more generally $H_m < \infty$. 13	Prove primes contain k-term APs for all k; count solutions to many finite-complexity linear systems in primes. 54
Correlation order	Essentially 2-point / second-moment detection (force ≥ 2 primes among shifts) using quadratic optimization in weights. 13	k-point / higher-order correlations controlled by U^{k-1} norms; obstruction class = nilsequences. 55
Main tools	Selberg sieve, GPY framework, Maynard sieve weights, admissible tuples, explicit numerical optimization; distribution in APs (BV and refinements). 4	W-trick; pseudorandom majorant ν ; transference/relative Szemerédi; Gowers norms; inverse theorems; nilsequence equidistribution; Möbius orthogonality. 15
Typical analytic inputs	Bombieri–Vinogradov level distribution; stronger/weighted variants improve constants; admissible tuple diameter computations matter. 56	Linear forms estimates for the majorant; structure-vs-randomness via GI(s) and MN(s) machinery (now theorems). 57
Representative unconditional quantitative results	$H_1 \leq 246$ (Polymath8b); Maynard $H_1 \leq 600$; Zhang $H_1 < 7 \times 10^7$. 58	Existence of k-term APs in primes for all k (Annals 2008). 59
Representative conditional bounds	$H_1 \leq 12$ under EH (Maynard); $H_1 \leq 6$ under generalized EH (Polymath8b). 60	Conditional asymptotics for wider pattern families framed in “Linear equations in primes” become unconditional after GI(s)/MN(s) theorems. 57
Recent/modern frontier (post-2015)	Bounded gaps in short intervals; distribution to large moduli with well-factorable weights; applications to prime clusters and related questions. 61	Higher-order uniformity in short intervals on average; quantitative inverse theorems; polynomial patterns and densification improvements to transference. 62

Attribute	Bounded-gaps program (Maynard–Tao / Polymath8b lineage)	Long-AP / linear-pattern program (Green–Tao lineage)
Key open problems	Twin primes (prove $H_1 = 2$); any major improvement in level of distribution beyond current reach; turn “some bounded gap” into “specific gap.” ⁶³	Strong quantitative bounds (effective $N(k)$); extend deeper into degenerate/ binary configurations (where affine dependence dominates). ⁶⁴

Recommended research experiments and computations

These are designed as **bridge tests**: they ask whether a proposed hybrid signal is “real structure” or merely “finite sieve coupling,” and they can be run at scales accessible to computational experimentation (e.g., primes up to 10^6 – 10^7 , depending on implementation).

Experiment	Purpose	Parameters	Data sources	Metrics	Expected outcome	Main pitfalls
Finite-sieve vs primes for 3-term AP counts after W-trick	Test whether observed AP suppression/ structure is explained by finite coupled sieving (as a majorant model)	$N \in \{2 \cdot 10^5, 10^6, 5 \cdot 10^6\}$; $W \in \{30, 210\}$; sieve limit B ladder (e.g. 31, 101, 317, 1009, ...)	Prime tables; constructed sieve survivors	Ratio of 3AP counts (primes / model), per residue class; convergence in B	Ratios stabilize toward prime data as B grows if “structure = sieve coupling”	Need correct normalization by local factors; boundary effects in counting
“3-point Maynard weight” toy optimization	Explore whether Maynard-style quadratic optimization can be sensibly extended to triple-pattern detectors in finite ranges	Fix pattern family (e.g. $(n, n + d, n + 2d)$ with $d \equiv 0 \pmod{W}$); discretize coefficient family	Synthetic divisor-sum weights + primes	Improvement of triple-hit rate vs baseline; compare to random/ sieve models	Likely modest gains that mirror sieve constraints; may reveal where parity barrier enters	Optimization landscape large; results may be dominated by small primes without careful W-pre-sieving

Experiment	Purpose	Parameters	Data sources	Metrics	Expected outcome	Main pitfalls
Local Fourier / U^2 diagnostics of primes vs sieve majorants in short windows	Compare “local pseudorandomness” diagnostics central to transference with sieve predictions	Window lengths H in $[N^\theta, N]$ for $\theta \in \{0.1, 0.2, 0.3\}$; average over x	Primes; sieve majorants	Max nontrivial Fourier coefficient; local U^2 norms averaged over x	Primes should sit close to sieve majorant baseline after W-trick; residual should be small on average	Window endpoints; need robust averaging; computational cost
Short-interval bounded-gaps frequency vs local uniformity proxies	Connect bounded gaps in $[x - x^\delta, x]$ to local uniformity signatures	$\delta \in [0.55, 1]$ and various x ranges	Primes	Frequency of gaps $\leq H$ within windows vs local spectral peaks	Hypothesis: stronger short-interval uniformity correlates with more “expected” bounded-gap frequencies	Correlation may be weak/noisy; requires careful conditioning on local density

Rationale and expected signature. If a “unified framework” is real rather than rhetorical, the sieve-majorant model should explain most low-complexity deviations (as in many Green-Tao-style pseudorandom majorants), while residual deviations should correlate with the precise obstruction classes predicted by inverse theorems (nilsequence-like structure). ⁶⁵

Notes on sources and how to read the frontier quickly

A high-yield reading path that stays close to primary sources is:

- Polymath8b paper: “Variants of the Selberg sieve, and bounded intervals containing many primes” (for the full bounded-gaps state of the art and sieve-theoretic limitations). ⁸
- Maynard Annals 2015 (clean GPY refinement and the core variational structure). ⁶⁶
- Green-Tao Annals 2008 (the original transference blueprint: W-trick, majorant, relative Szemerédi). ⁵⁹
- Green-Tao-Ziegler inverse theorem (what “structure” means in higher-order uniformity). ⁶⁷
- Green-Tao Möbius vs nilsequences (how analytic number theory kills those structures). ⁵¹
- Annals 2023 “Higher uniformity ... in short intervals” (modern local uniformity bridge; shows nilsequence tests in short intervals on average). ⁴⁵
- Maynard 2020 “Primes in AP to large moduli II” (distribution frontier feeding sieve weights). ³⁴

This collection, taken together, effectively answers the “can they be unified?” question: they share local-factor removal and sieve-weight DNA, but they diverge sharply in which correlations they must control and what “structured obstructions” look like at the required depth. ⁶⁸

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- ¹ ³ ²⁴ ⁵⁵ ⁶⁷ **An inverse theorem for the Gowers $U^{s+1}[N]$ -norm**
https://annals.math.princeton.edu/wp-content/uploads/annals-v176-n2-p11-p.pdf?utm_source=chatgpt.com
- ² ⁷ ³² ⁴⁰ ⁴⁹ ⁶¹ **Bounded gaps between primes in short intervals**
https://arxiv.org/abs/1707.05437?utm_source=chatgpt.com
- ⁴ ¹¹ ¹³ ¹⁴ ¹⁹ ²⁸ ⁴³ ⁶⁶ **Small gaps between primes - Annals of Mathematics**
https://annals.math.princeton.edu/2015/181-1/p07?utm_source=chatgpt.com
- ⁵ ¹⁰ ¹² ¹⁵ ²³ ⁴² ⁵⁴ ⁵⁹ ⁶⁵ **The primes contain arbitrarily long arithmetic progressions**
https://annals.math.princeton.edu/wp-content/uploads/annals-v167-n2-p03.pdf?utm_source=chatgpt.com
- ⁶ ¹⁷ ²² ²⁶ ³¹ ⁶⁸ **The primes contain arbitrarily long arithmetic progressions**
https://arxiv.org/abs/math/0404188?utm_source=chatgpt.com
- ⁸ ¹⁸ ²¹ ²⁹ ³⁰ ³⁶ ⁵⁸ ⁶³ **Variants of the Selberg sieve, and bounded intervals containing many primes**
https://arxiv.org/abs/1407.4897?utm_source=chatgpt.com
- ⁹ **(PDF) A Weighted Distribution of Primes and a New ...**
https://www.researchgate.net/publication/393888742_A_Weighted_Distribution_of_Primes_and_a_New_Unconditional_Bound_on_Gaps_Between_Primes?utm_source=chatgpt.com
- ¹⁶ ⁵⁰ ⁵⁷ ⁶⁴ **Linear Equations in Primes**
https://arxiv.org/abs/math/0606088?utm_source=chatgpt.com
- ²⁰ ²⁷ ⁵⁶ **Bounded gaps between primes - Annals of Mathematics**
https://annals.math.princeton.edu/2014/179-3/p07?utm_source=chatgpt.com
- ²⁵ ⁴¹ ⁵¹ **The Möbius function is strongly orthogonal to nilsequences**
https://annals.math.princeton.edu/wp-content/uploads/annals-v175-n2-p03-p.pdf?utm_source=chatgpt.com
- ³³ ⁴⁴ **[1910.13450] Gaps between primes**
https://arxiv.org/abs/1910.13450?utm_source=chatgpt.com
- ³⁴ ³⁹ **Primes in arithmetic progressions to large moduli II**
https://arxiv.org/abs/2006.07088?utm_source=chatgpt.com
- ³⁵ **Linear equations in primes - Annals of Mathematics**
https://annals.math.princeton.edu/2010/171-3/p08?utm_source=chatgpt.com
- ³⁷ **Quantitative bounds in the inverse theorem for the Gowers U^{s+1} -norms over cyclic groups**
https://arxiv.org/abs/1811.00718?utm_source=chatgpt.com
- ³⁸ **Séminaire BOURBAKI Mars 2021 73e ...**
https://www.bourbaki.fr/TEXTES/Exp1174-Bloom.pdf?utm_source=chatgpt.com
- ⁴⁵ ⁶² **Higher uniformity of bounded multiplicative functions in short ...**
https://annals.math.princeton.edu/2023/197-2/p03?utm_source=chatgpt.com

46 52 53 Higher uniformity of bounded multiplicative functions in short intervals on average

https://arxiv.org/abs/2007.15644?utm_source=chatgpt.com

47 Multiplicative functions in short intervals - Annals of Mathematics

https://annals.math.princeton.edu/wp-content/uploads/annals-v183-n3-p06-p.pdf?utm_source=chatgpt.com

48 Higher uniformity of bounded multiplicative functions in short ...

https://terrytao.wordpress.com/2020/07/30/higher-uniformity-of-bounded-multiplicative-functions-in-short-intervals-on-average/?utm_source=chatgpt.com

60 Small gaps between primes

https://arxiv.org/abs/1311.4600?utm_source=chatgpt.com